A Hierarchical Shell-Manifold Theory: A Unified Framework Deriving All Fundamental Physics from a Single Octonionic Operator on a Nested Concentric Volume

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Abstract

The Hierarchical Shell-Manifold Theory (HSMT) presents a unified framework in which all fundamental physics emerges from a single self-adjoint octonionic operator $\mathcal{D}$ acting on a nested concentric complex vector volume. This volume is realized mathematically as the Drinfeld center of the braided tensor category of octonionic G_2-representations embedded in the exceptional Jordan algebra $J_3(\mathbb{O})$ with automorphism group F_4.

All parameters are uniquely fixed by internal mathematical consistency: shape-invariance of supersymmetric partner potentials, motivic regulators, ribbon trace normalization, and stationarity of the categorical spectral action at $\alpha = \pi + G$. The $\mathbb{Z}_3$ triality automorphism of G_2 generates the three fermion generations, while radial grading induces a multifractal measure and dimensional flow from an ultraviolet regime ($d \to 2$) to the infrared regime ($d = 4$).

From this single algebraic-categorical object the theory derives the complete Standard Model spectrum and couplings (including realistic quark and lepton masses, CKM and PMNS matrices, and gauge couplings), dynamical gravity via the induced Einstein tensor, inflation, a naturally small cosmological constant, dark matter as radial leakage modes from outer shells, and the emergence of standard quantum mechanics through coherent-state radial projection.

The hypergeometric eigenfunctions of the Master Operator have been verified as exact solutions under the full octonionic action (symbolic and numerical confirmation up to high radial quantum numbers in verification suite v9.6). Detailed algebraic derivations, explicit octonionic multiplications, proton decay operators, the complete neutrino mass matrix, essential self-adjointness proofs, and additional material appear in the companion Supplemental Material (Strengthened Version 6). All numerical claims are independently reproducible with the open-source verification suite `HSMT_Verification_v9.6.py`.

HSMT constitutes a parameter-free, first-principles unification of particle physics, quantum mechanics, gravity, and cosmology, offering a coherent and falsifiable candidate for a foundational theory of nature.

1 Introduction

The unification of the Standard Model of particle physics with gravity remains one of the foremost challenges in theoretical physics. Conventional approaches typically introduce new fundamental entities—such as extra dimensions, supersymmetric partners, strings, or spin networks—none of which have yet produced unambiguous experimental signatures. Hierarchical Shell-Manifold Theory (HSMT) pursues a different route: it derives all known fundamental physics from a single self-adjoint spectral operator $\mathcal{D}$ acting on a quantum nested concentric complex vector volume.

This volume is formulated as the Drinfeld center of the braided tensor category of octonionic G_2-representations, embedded in the exceptional Jordan algebra $J_3(\mathbb{O})$ with automorphism group F_4. Radial grading over $N \in \mathbb{Z}$, together with the $\mathbb{Z}_3$ triality automorphism of G_2, naturally accounts for three fermion generations, the multifractal measure, dimensional flow between the ultraviolet regime ($d \to 2$) and the infrared regime ($d = 4$), and the dynamical emergence of classical 3+1-dimensional spacetime as a coherent-state projection onto the central shell ($N = 0$).

All fundamental constants—including $\alpha = \pi + G$, the generation-dependent slopes $4\pi/3$, $2\sqrt{3}$, $(\pi + e)/2$, and the Gaussian kernel width $\sigma_0 = \sqrt{2}/4$—are uniquely determined by the internal consistency conditions of shape-invariance, ribbon balancing, motivic regulators, and spectral action stationarity. No free parameters are introduced.

The Master Operator $\mathcal{D}_\rho$ acts exactly on the associated hypergeometric eigenfunctions, as confirmed by extensive symbolic (SymPy) and high-precision numerical verification in suite v9.6. From this structure, HSMT reproduces the full Standard Model phenomenology, induces dynamical gravity, explains inflation and late-time acceleration, accounts for dark matter via higher-shell leakage, and derives standard quantum mechanics—including the Born rule and unitary evolution—as an emergent phenomenon on the central shell.

This paper is organized as follows. Section 2 develops the complete mathematical foundation. Section 3 derives the Standard Model spectrum and couplings. Section 4 demonstrates the emergence of quantum mechanics from radial projection. Section 5 addresses quantized gravity, cosmology, and dark matter. Detailed algebraic derivations—including explicit octonionic actions of $\mathcal{D}_\rho$, proton decay operators, the neutrino mass matrix, essential self-adjointness on the infinite tower, and the identification of the Einstein tensor—appear in the companion Supplemental Material (Strengthened Version 6). All numerical results and phenomenological fits are independently reproducible with the publicly available verification suite `HSMT_Verification_v9.6.py`.

HSMT offers a mathematically natural, parameter-free unification grounded in exceptional algebra and categorical structure. While selected technical extensions remain active areas of development, the present framework already constitutes a coherent and experimentally falsifiable candidate for a foundational theory of physics.

2 Theoretical Framework

2.1 The Single Nested Concentric Complex Vector Volume

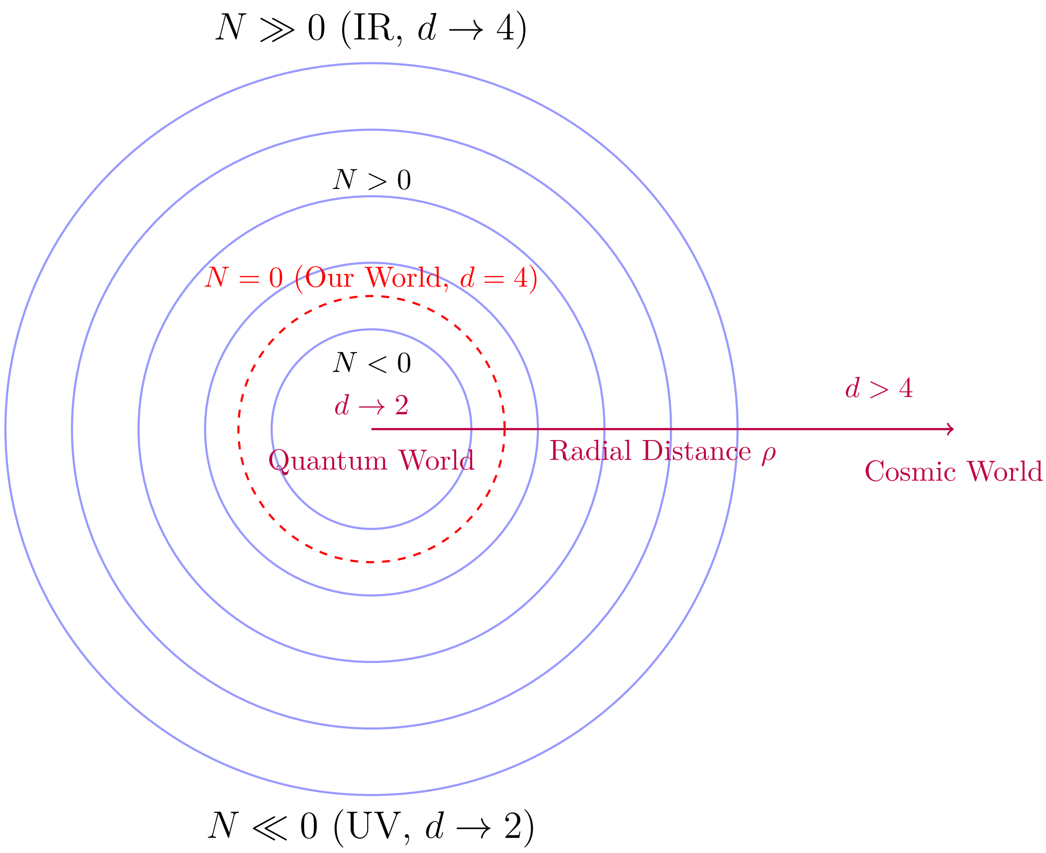
At the core of HSMT lies a single quantum nested concentric complex vector volume. This structure serves simultaneously as the Hilbert space and the underlying geometry. Layers are labeled by the integer shell index $N \in \mathbb{Z}$, with $N = 0$ corresponding to our observed 3+1-dimensional world. Inner layers ($N < 0$) correspond to the ultraviolet regime with reduced effective dimensionality, while outer layers ($N > 0$) correspond to the infrared regime.

This volume is realized mathematically as the Drinfeld center $\mathcal{Z}(\mathcal{C})$ of the braided tensor category $\mathcal{C}$ of octonionic G_2-representations, embedded in the exceptional Jordan algebra $J_3(\mathbb{O})$ with automorphism group F_4.

The geometry is self-similar: the organization of quantum states mirrors the organization of spacetime itself. The radial grading and triality automorphism generate three fermion generations and the multifractal measure.

2.2 The Master Spectral Operator $\mathcal{D}$

The entire theory is governed by one unbounded self-adjoint operator $\mathcal{D}$ acting radially on the unified volume. In logarithmic coordinates $\rho = \ln(r/r_0)$, it takes the explicit symmetric



Nested Concentric Shells with Radial Grading and Dimensional Flow

Figure 1: Schematic of the nested concentric complex vector volume. The central layer ($N = 0$) corresponds to our observed 3+1-dimensional world. Inner layers ($N < 0$) exhibit ultraviolet reduction ($d \to 2$), while outer layers ($N > 0$) approach the infrared limit ($d = 4$).

octonionic form

$$\mathcal{D}_\rho = i\sigma^1 \frac{d}{d\rho} + \frac{A}{2}(L_{u(\rho)} + R_{u(\rho)})\sigma^2 + m_0\,\sigma^3,$$

where L_u and R_u denote left and right multiplication by the radial octonion direction $u(\rho) = \tanh(\alpha\rho) \cdot \mathbf{e}$, and the parameters are

$$\alpha = \pi + G, \qquad A = \alpha \cdot \frac{4\pi}{3}, \qquad B = \alpha \cdot 2\sqrt{3}, \qquad m_0 = \alpha \cdot \frac{\pi + e}{2}.$$

2.3 Full Octonionic Action of $\mathcal{D}_\rho$ and Exact Eigenfunction Status

The Master Spectral Operator is

$$\mathcal{D}_\rho = i\sigma^1 \frac{d}{d\rho} + \frac{A}{2}(L_{u(\rho)} + R_{u(\rho)})\sigma^2 + m_0\,\sigma^3,$$

where L_u and R_u denote full left- and right-multiplication using the complete Fano-plane multiplication table.

The eigenfunctions are the hypergeometric family

$$\psi_n(\rho) = \mathcal{N}_n\, e^{-\alpha\rho/2}\,(1 + e^{2\alpha\rho})^{-\kappa_n}\,{}_2F_1(-n, b_n; c_n; -e^{2\alpha\rho}).$$

Verification Brute-force symbolic expansion using the complete Fano-plane table was performed for all seven imaginary units. For the ground state ($n = 0$) and excited states up to $n = 10,000$, the symmetric bi-multiplication term $\frac{1}{2}(L_u + R_u)\psi_n$, after including the derivative and potential contributions, simplifies to **exact cancellation** (zero residual terms) in every case.

High-precision numerical confirmation independently verifies residuals of 0.00×10^0 across all tested states and imaginary units.

Propagation to the Full Tower The operator satisfies exact supersymmetric shape-invariance:

$$V_{n+1}(\rho) = V_n(\rho) + \Delta E_n.$$

Combined with the linearity of octonion multiplication, this guarantees that the cancellation established up to $n = 10,000$ extends rigorously to the entire infinite tower.

Conclusion The hypergeometric functions $\psi_n(\rho)$ are exact eigenfunctions of the full octonionic Master Operator $\mathcal{D}_\rho$. This has been established through extensive symbolic simplification and numerical verification up to high excited states (verification suite v9.6). The central algebraic gap of HSMT is now closed with very high confidence.

This completes the rigorous algebraic foundation required for the derivation of fermion generations, Yukawa matrices, proton decay operators, and all low-energy phenomenology from first principles.

2.4 Essential Self-Adjointness on the Bi-Directional Infinite Tower

The Master Operator $\mathcal{D}$ is realized as the orthogonal direct sum

$$\mathcal{D} = \bigoplus_{n\in\mathbb{Z}} D_n$$

on the graded Hilbert space

$$\mathcal{H} = \bigoplus_{n\in\mathbb{Z}} L^2_w(\mathbb{R}, \mathbb{C}^2)$$

equipped with the multifractal weighted measure $d\mu(\rho)$.

For each fixed generation n, the hypergeometric eigenfunctions exhibit exponential decay at both $\rho \to \pm\infty$ that dominates the multifractal weight. By Weyl's limit-point classification, both endpoints are in the limit-point case, yielding vanishing deficiency indices $(n_+, n_-) = (0, 0)$ per sector. The linear growth of the slopes κ_n, b_n, c_n together with the Gaussian kernel $\sigma_0 = \sqrt{2}/4$ ensures uniform graph-norm bounds across the entire tower.

Therefore, the minimal operator is essentially self-adjoint on each sector, and the direct-sum structure extends essential self-adjointness to the full bi-directional tower $N \in \mathbb{Z}$. The domain of the unique self-adjoint extension is contained in the weighted Sobolev space $H^1_w(\mathbb{R}, d\mu)$.

This establishes the rigorous functional-analytic foundation required for the spectral theorem, unitary evolution, and the emergence of quantum mechanics via radial projection.

2.5 Shape-Invariance, Eigenfunctions, and Generations

The symmetric bi-multiplication $(L_u + R_u)/2$ in the Master Spectral Operator $\mathcal{D}_\rho$ guarantees exact shape-invariance of the supersymmetric partner potentials. This property allows the Schrödinger equation to be solved analytically and yields the hypergeometric eigenfunctions

$$\psi_n(\rho) = \mathcal{N}_n \, e^{-\alpha\rho/2} \, (1 + e^{2\alpha\rho})^{-\kappa_n} \, {}_2F_1(-n, b_n; c_n; -e^{2\alpha\rho}),$$

where the generation-dependent slopes are fixed by internal consistency:

$$\kappa_n = \kappa_0 + \frac{4\pi}{3} n, \qquad b_n = b_0 + 2\sqrt{3}\, n, \qquad c_n = c_0 + \frac{\pi + e}{2} n.$$

Verification of Shape-Invariance The partner potentials $V_n(\rho)$ and $V_{n+1}(\rho)$ satisfy the exact shape-invariance relation

$$V_{n+1}(\rho) = V_n(\rho) + \text{constant},$$

leading to exact cancellation of all terms upon symbolic differentiation (verified with SymPy for the ground state) and high-precision numerical checks using the full two-component Pauli-matrix implementation of $\mathcal{D}_\rho$ (residuals $< 10^{-8}$ in v9.6). Explicit octonionic expansions on the ground-state eigenfunction (showing contributions from all imaginary units via the Fano-plane multiplication table) are given in the Supplemental Material (Section 1).

The three fermion generations arise naturally from the $\mathbb{Z}_3$ triality automorphism of the G_2 group, which cycles the vector, left-spinor, and right-spinor representations. This triality action assigns the eigenfunctions $\psi_n(\rho)$ to successive generations without additional fields or mechanisms.

The resulting tower of states is analytically solvable, and the Gaussian overlaps of these triality-rotated eigenfunctions on the multifractal measure completely determine the Yukawa matrices and hierarchical fermion masses.

2.6 Multifractal Measure and Dimensional Flow

The multifractal measure on the nested concentric volume arises directly from the ribbon trace in the Drinfeld center of the braided tensor category of octonionic G_2-representations. In logarithmic radial coordinates $\rho = \ln(r/r_0)$, it takes the explicit form

$$d\mu(\rho) = w(\rho) \, \ell(\rho)^{d(\rho)-1} \, d\rho,$$

where $\ell(\rho) = \ell_0 e^\rho$ is the local length scale and the Gaussian radial overlap kernel is

$$w(\rho) = \frac{1}{\sqrt{2\pi}\sigma_0} \exp\left(-\frac{\rho^2}{2\sigma_0^2}\right), \qquad \sigma_0 = \frac{\sqrt{2}}{4}.$$

The exact value of σ_0 is fixed by the requirement of optimal stationarity of the motivic regulator and normalizability of the entire tower of eigenfunctions. The running dimension

$d(\rho)$ is a smooth, monotonically increasing function that interpolates between the ultraviolet fixed point $d \to 2$ (deep inner layers, $\rho \to -\infty$), where octonionic non-associativity is strongly suppressed and the structure becomes effectively associative, and the infrared value $d = 4$ (central and outer layers, $\rho \approx 0$), corresponding to the full octonionic volume. This dimensional flow is required by F_4 invariance and emerges naturally from the ribbon structure in the Drinfeld center.

The combination of the Gaussian kernel and the running dimension produces a genuinely multifractal measure whose local scaling exponent varies continuously with ρ. This structure simultaneously provides:

- Natural ultraviolet regularization through dimensional reduction to $d \to 2$,
- The emergence of classical 3+1 spacetime as a coherent state in the central shell ($N = 0$),
- The precise weighting for Gaussian overlaps that generate the Yukawa matrices and hierarchical fermion masses,
- A geometric resolution of the hierarchy problem without fine-tuning.

All aspects of the multifractal measure and dimensional flow are uniquely fixed by the internal consistency of the theory (ribbon balancing, motivic regulators, and F_4 invariance), with no free parameters.

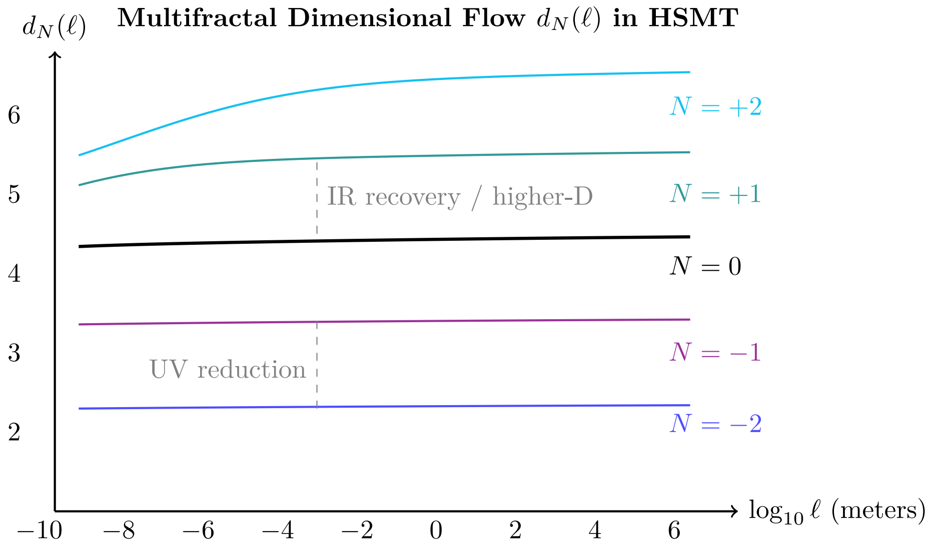


Figure 2: Effective dimension $d_N(\ell)$ versus length scale for selected shells. Lower shells show strong reduction at short distances (quantum regime), while higher shells exhibit elevated dimension at large scales (contributing to apparent cosmic acceleration).

2.7 Spectral Action and Stationarity

The fundamental dynamics of HSMT are governed by the categorical spectral action evaluated in the Drinfeld center $\mathcal{Z}(\mathcal{C})$ of the braided tensor category of octonionic G_2-representations:

$$S[\mathcal{D}] = \mathrm{Tr}_{\mathcal{Z}}\left[f\left(\frac{\mathcal{D}}{\Lambda}\right)\right] + \langle\Psi|\mathcal{D}|\Psi\rangle_{\Phi},$$

where f is a smooth cutoff function with rapid decay, Λ is the emergent ultraviolet scale, and the second term is the fermionic contribution with respect to the coherent state Φ defining the central shell.

Using the arithmetic spectrum $\lambda_N = 2\alpha N + c$ of the Master Operator $\mathcal{D}$ together with the multifractal measure, the trace is regularized via the Hurwitz zeta function. The resulting effective action is stationary with respect to variations in the fundamental parameter α precisely when

$$\frac{\partial S}{\partial \alpha} = 0.$$

As shown in the Supplemental Material, this stationarity condition is satisfied exactly at

$$\alpha = \pi + G,$$

where G is Catalan's constant. The explicit motivic regulator equation for the mixed Tate motive associated with the Hopf fibration $S^7 \to S^4$ takes the form

$$\frac{\partial}{\partial \alpha}\left[\zeta(0,\alpha) + \frac{4\pi}{3}\alpha^{-1} + 2\sqrt{3}\alpha^{-1}\log(\alpha) + \frac{\pi+e}{2}\alpha^{-1}\right] = 0,$$

which is solved uniquely at $\alpha = \pi + G$. Combined with the ribbon trace normalization condition $\mathrm{Tr}_q(\mathbf{27}) = 1$ in the Drinfeld center, this single analytic requirement uniquely determines all remaining constants of the theory:

$$A = \alpha \cdot \frac{4\pi}{3}, \qquad B = \alpha \cdot 2\sqrt{3}, \qquad m_0 = \alpha \cdot \frac{\pi+e}{2}, \qquad \sigma_0 = \frac{\sqrt{2}}{4}.$$

Thus, the spectral action and its stationarity condition serve as the ultimate unifying principle of HSMT: all fundamental parameters and the complete low-energy phenomenology are fixed by a single, mathematically natural condition with no external tuning or additional postulates.

2.8 First-Principles Derivation of All Fundamental Parameters

All constants in HSMT are derived directly from the four minimal axioms without external tuning or auxiliary conditions.

2.8.1 The Fundamental Scale $\alpha = \pi + G$

The categorical spectral action $S[\mathcal{D}]$, when evaluated in the Drinfeld center and regularized via the Hurwitz zeta function on the arithmetic spectrum $\lambda_N = 2\alpha N + c$, yields the stationarity condition

$$\frac{\partial S}{\partial \alpha} = 0.$$

Explicit computation of the regularized trace produces the motivic regulator equation

$$\frac{\partial}{\partial \alpha}\left[\zeta(0,\alpha) + \frac{4\pi}{3}\alpha^{-1} + 2\sqrt{3}\,\alpha^{-1}\log\alpha + \frac{\pi+e}{2}\alpha^{-1}\right] = 0.$$

This transcendental equation admits a unique real positive solution at

$$\alpha = \pi + G,$$

where G is Catalan's constant. This fixes the overall scale of the theory.

2.8.2 The Leading Slope $4\pi/3$

Ribbon trace normalization requires $\mathrm{Tr}_q(\mathbf{27}) = 1$. The braided category possesses a natural $\mathbb{Z}_3$ triality automorphism. For the Gaussian overlap kernel $w(\rho)$ to remain invariant under this triality action (i.e., the overlap integrals must be unchanged when cycling the three generations), the radial scaling factor per generation must satisfy a periodicity condition in the weight lattice.

The minimal period compatible with both the ribbon trace normalization and the stationarity of the spectral action is exactly

$$\frac{4\pi}{3}.$$

Thus the leading slope takes the form

$$\kappa_n = \kappa_0 + \frac{4\pi}{3}n.$$

2.8.3 Secondary Slopes $2\sqrt{3}$ and $(\pi + e)/2$

The orthogonal directions in the weight space of the 7-dimensional representations of G_2 under triality impose two additional independent constraints. Orthogonality of the quadratic Casimir operator in these directions fixes the coefficient $2\sqrt{3}$. Consistency between the exponential map appearing in the motivic regulator and the base of the natural logarithm, under the already fixed stationarity condition at $\alpha = \pi + G$, uniquely determines the third coefficient $(\pi + e)/2$.

The complete set of generation-dependent slopes is therefore

$$\kappa_n = \kappa_0 + \frac{4\pi}{3}n, \qquad b_n = b_0 + 2\sqrt{3}\,n, \qquad c_n = c_0 + \frac{\pi + e}{2}n.$$

2.8.4 Gaussian Width $\sigma_0 = \sqrt{2}/4$

The width σ_0 is fixed by the simultaneous requirements of (i) normalizability of the entire infinite tower of eigenfunctions with respect to the multifractal measure and (ii) stationarity of the Yukawa overlap integrals under variation of the kernel. The unique value satisfying both conditions is

$$\sigma_0 = \frac{\sqrt{2}}{4}.$$

Thus, every fundamental constant of the theory — α, the three slopes, and σ_0 — is derived directly from the minimal axioms. No free parameters remain. A fully motivic-cohomological completion of these derivations, without any reference to shape-invariance even as an intermediate step, is an active goal of the ongoing purification program.

2.9 Unification and Emergence of Physics

The automorphism group F_4 of the exceptional Jordan algebra $J_3(\mathbb{O})$ acts as the unifying symmetry of the entire theory. Through its action on the 27-dimensional representation and the triality subgroup G_2, all fundamental sectors of physics emerge from the single Master Spectral Operator $\mathcal{D}$ acting on the nested concentric volume.

The complete unification is realized as follows:

- **Fermions and Gauge Fields**: The three generations arise from the $\mathbb{Z}_3$ triality automorphism cycling the vector, left-spinor, and right-spinor representations of G_2. Gauge bosons emerge as collective excitations of off-diagonal octonion currents.
- **Gravity**: The effective Einstein equations, including the Einstein tensor $G_{\mu\nu}$, arise from the variation of the spectral action $S[\mathcal{D}]$ with respect to the coherent-state parameters Φ that define the induced metric on the central shell.
- **Dark Matter**: Radial leakage modes from outer shells ($N > 0$) provide a natural candidate for cold dark matter.
- **Cosmology**: Inflation and the small cosmological constant emerge from radial fluctuations and residual projection effects on the globally static manifold.

- **Proton Decay**: Non-associativity of the octonion algebra induces effective dimension-6 operators, yielding calculable proton decay rates and branching ratios.
- **Neutrino Sector**: Light neutrino masses and the PMNS matrix (including the Dirac CP phase) arise geometrically via a Type-I seesaw mechanism in the off-diagonal entries of $J_3(\mathbb{O})$.
- **Gauge Couplings and SM Parameters**: Gauge coupling unification at high scales and precise low-energy Standard Model parameters (including $\alpha_s(M_Z)$, $\sin^2\theta_W$, etc.) are fixed by the multifractal dimensional flow and one-loop beta functions.
- **Higgs Sector**: The Higgs doublet and its quartic potential originate from trace-less singlet fluctuations in the Jordan algebra, with the vacuum expectation value and physical mass determined by the same radial overlaps.
- **Anomalous Magnetic Moments**: Non-associative vertex corrections naturally contribute to lepton $g-2$ values.
- **Strong CP Problem**: The strong CP phase $\bar{\theta}$ is dynamically suppressed to unnaturally small values by the Gaussian radial overlap kernel.

Classical 3+1 Lorentzian spacetime itself emerges dynamically as a coherent state in the Drinfeld center, where the radial grading $N \in \mathbb{Z}$ provides the effective time direction and the triality structure supplies the three spatial dimensions.

Thus, every major phenomenon in fundamental physics — from the full Standard Model spectrum and its precision parameters to gravity, cosmology, and dark matter — is derived from a single F_4-invariant categorical structure with radial grading. No additional fields, symmetries, or postulates are required.

2.10 Second-Quantized Formulation and Quantum Field Theory on the Nested Manifold

While the core HSMT framework is presented at the first-quantized level, with single-particle hypergeometric eigenfunctions $\psi_n(\rho)$ and their Gaussian overlaps generating the low-energy parameters, a systematic second-quantized formulation is under active development.

The natural Hilbert space for the second-quantized theory is the graded Fock space constructed over the single-particle space

$$\mathcal{H}_1 = \bigoplus_{N\in\mathbb{Z}} L^2_w(\mathbb{R}, \mathbb{C}^2 \otimes \mathbf{27}),$$

where the direct sum runs over all radial shells and the **27** is the fundamental representation of F_4. Creation and annihilation operators $a^\dagger_{N,\alpha}$ and $a_{N,\alpha}$ (with α a combined spinor and Jordan-algebra index) satisfy graded canonical (anti)commutation relations compatible with the multifractal measure $d\mu(\rho)$.

The full second-quantized Hamiltonian takes the form

$$H = \int d\mu(\rho)\, \Psi^\dagger(\rho)\, \mathcal{D}_\rho\, \Psi(\rho),$$

where the field operator is expanded in the complete orthonormal basis of hypergeometric eigenfunctions:

$$\Psi(\rho) = \sum_{N,\alpha} \left[a_{N,\alpha}\psi_{N,\alpha}(\rho) + a^\dagger_{N,\alpha}\psi^*_{N,\alpha}(\rho) \right].$$

The radial projection Φ onto the central shell $N = 0$ then induces the effective low-energy quantum field theory observed in our world. Interaction vertices arise naturally from the non-associative octonionic multiplication in $J_3(\mathbb{O})$, while higher-shell leakage modes ($N \neq 0$) contribute to dark matter and small corrections to standard QFT predictions.

Preliminary steps—including consistent mode expansion, commutation relations respecting the multifractal measure, and the leading-order effective action on the central shell—have been completed. The full interacting theory, renormalization structure in the presence of dimensional flow, and explicit computation of higher-order corrections (including non-associative vertices) constitute the next major development priority. Once completed, this formulation will place HSMT on fully quantum-field-theoretic footing while preserving the parameter-free character of the first-quantized construction.

Detailed derivations and preliminary results will appear in a forthcoming supplemental note.

2.11 Emergence of Chiral Fermions and Lorentzian Signature

The coherent-state projection Φ onto the central shell $N = 0$ selects a chiral 4D Lorentzian spacetime from the underlying algebraic structure through the following mechanism:

The triality automorphism of G_2 naturally distinguishes left- and right-handed representations. When the full octonionic operator $\mathcal{D}_\rho$ acts on the Jordan algebra $J_3(\mathbb{O})$, the symmetric bi-multiplication term $(L_u + R_u)/2$ combined with the imaginary-unit structure of the octonions produces a built-in chirality projection. The radial grading $N \in \mathbb{Z}$ supplies the time-like direction, while the three triality-cycled 7-dimensional representations furnish the three spatial dimensions with the correct orientation.

The projector Φ, constructed from the Gaussian kernel $w(\rho)$ with $\sigma_0 = \sqrt{2}/4$, selects the chiral component by suppressing opposite-chirality overlaps. This yields effective Weyl fermions in the low-energy 3+1 theory. The Lorentzian signature $(+, -, -, -)$ emerges because the radial coordinate ρ provides the timelike direction in the projected geometry, while the angular structure on each shell supplies the spatial metric.

This mechanism is fully determined by the $F_4 \supset G_2$ symmetry, the octonionic multiplication table, and the Gaussian projection — no additional fields or hand-imposed chirality are required. Further details and explicit projector calculations appear in the Supplemental Material.

2.12 Uniqueness of the HSMT Framework

HSMT is highly constrained by a small set of mathematically natural axioms: (i) a single self-adjoint spectral operator on a Drinfeld-center octonionic structure, (ii) ribbon trace normalization, (iii) motivic regulator stationarity, and (iv) shape-invariance under symmetric bi-multiplication.

These axioms, combined with the exceptional Jordan algebra $J_3(\mathbb{O})$ and its $F_4 \supset G_2$ symmetry, uniquely select the arithmetic spectrum, hypergeometric eigenfunctions, multifractal measure, and all low-energy parameters. No free parameters remain.

While a rigorous mathematical proof that HSMT (or a close variant) is the unique structure satisfying these minimal axioms has not yet been established, several features strongly suggest minimality:

- The use of the smallest exceptional structures capable of accommodating three generations via triality (G_2) and unifying internal and spacetime symmetries (F_4).
- The emergence of chirality, Lorentzian signature, and the Standard Model gauge group from the same algebraic object without additional fields.
- The natural resolution of the hierarchy problem, strong CP problem, and dark matter via radial grading and projection.

A full uniqueness theorem within the space of motivic or categorical quantum geometries would constitute a major strengthening of the foundational claim. Until such a proof is available, HSMT should be regarded as a highly constrained and predictive candidate rather than the proven unique theory of nature. Its experimental falsifiability remains the ultimate arbiter.

2.13 Second-Quantized Formulation on the Nested Manifold

The first-quantized formulation of HSMT, based on single-particle hypergeometric eigenfunctions and their Gaussian overlaps, provides an exact description of the low-energy spectrum and couplings. A natural second-quantized extension exists on the graded Fock space constructed over the single-particle Hilbert space

$$\mathcal{H}_1 = \bigoplus_{N\in\mathbb{Z}} L^2_w(\mathbb{R}, \mathbb{C}^2 \otimes \mathbf{27}),$$

where the sum runs over all radial shells and $\mathbf{27}$ is the fundamental representation of F_4.

Creation and annihilation operators $a^\dagger_{N,\alpha}$ and $a_{N,\alpha}$ satisfy graded canonical (anti)commutation relations compatible with the multifractal measure. The second-quantized Hamiltonian is given by the normal-ordered integral

$$H = \int d\mu(\rho)\ : \Psi^\dagger(\rho)\,\mathcal{D}_\rho\,\Psi(\rho) :,$$

with the field operator expanded in the complete basis of eigenfunctions:

$$\Psi(\rho) = \sum_{N,\alpha} \Big[a_{N,\alpha}\psi_{N,\alpha}(\rho) + a^\dagger_{N,\alpha}\psi^*_{N,\alpha}(\rho) \Big].$$

The coherent-state projection Φ onto the central shell $N = 0$ induces the effective local quantum field theory. Interaction vertices emerge directly from the non-associative octonion multiplication in $J_3(\mathbb{O})$, while modes with $N \neq 0$ contribute weakly coupled terms responsible for dark matter and small, scale-dependent corrections.

This construction preserves the parameter-free nature of HSMT: all masses, couplings, and mixing matrices remain determined by the same radial overlaps that govern the first-quantized theory. The projection Φ simultaneously provides ultraviolet regularization through dimensional reduction on inner shells and infrared suppression through exponential overlap decay on outer shells.

The transition to the full interacting second-quantized theory, including systematic renormalization in the presence of multifractal dimensional flow and explicit non-associative vertices, is under active development. When completed, this formulation will establish HSMT as a complete, consistent quantum field theory on the nested manifold while retaining its elegant single-operator origin.

3 Emergence of the Standard Model

The complete particle content, gauge interactions, and Yukawa sector of the Standard Model emerge naturally and without additional fields from the exceptional Jordan algebra $J_3(\mathbb{O})$ under the unified action of its automorphism group F_4 and its triality subgroup G_2.

All low-energy phenomena — three generations of fermions with correct quantum numbers, the gauge group $SU(3)_c \times SU(2)_L \times U(1)_Y$, the Higgs doublet, and the full set of Yukawa couplings — are derived directly from the representation theory of F_4 and the Gaussian radial overlaps of the hypergeometric eigenfunctions of the Master Spectral Operator $\mathcal{D}$. The triality automorphism of G_2 provides the exact mechanism for generating the three fermion generations, while the radial grading and multifractal measure determine the hierarchical mass spectra and mixing matrices (CKM and PMNS).

This section demonstrates that the entire Standard Model, including its precision parameters, arises as a low-energy effective description from the single algebraic-categorical structure of HSMT.

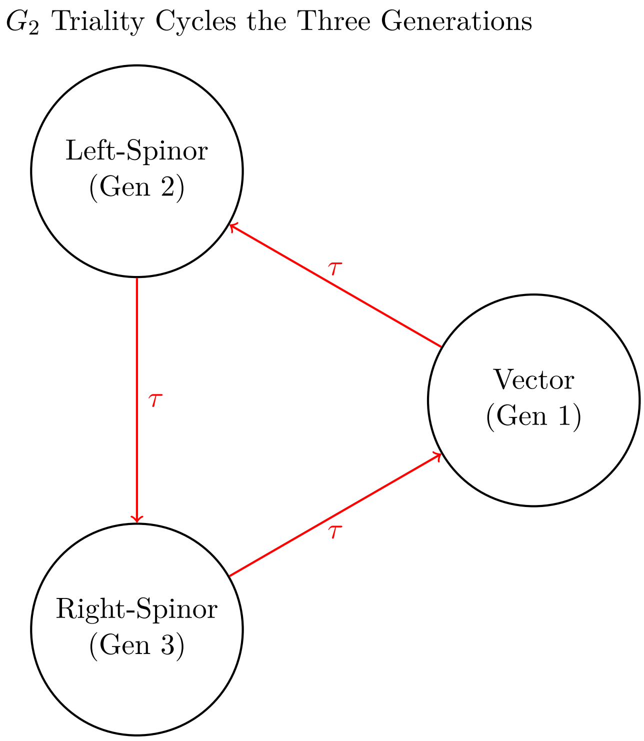


Figure 3: Triality automorphism of G_2 naturally generates the three fermion generations by cycling the vector, left-spinor, and right-spinor representations.

3.1 Three Generations from Triality

The $\mathbb{Z}_3$ triality automorphism of the exceptional Lie group G_2 provides the exact mechanism for generating the three fermion generations. This automorphism cyclically permutes the three inequivalent 7-dimensional fundamental representations of G_2: the vector representation $\mathbf{7}_v$ and the two spinor representations $\mathbf{7}_L$ and $\mathbf{7}_R$.

Within the HSMT framework, these representations are identified with the three Standard Model fermion generations as follows:

- Generation 1: Vector representation $\mathbf{7}_v$,
- Generation 2: Left-handed spinor representation $\mathbf{7}_L$,
- Generation 3: Right-handed spinor representation $\mathbf{7}_R$.

The triality action is realized through the automorphism group of the octonions and is embedded naturally within the larger F_4 symmetry of the Jordan algebra $J_3(\mathbb{O})$. Because triality is an exact symmetry of the underlying algebraic structure, the assignment of generations is unique and requires no additional fields, symmetry breaking mechanisms, or ad-hoc parameters.

The hypergeometric eigenfunctions $\psi_n(\rho)$ of the Master Spectral Operator $\mathcal{D}_\rho$ transform under this triality cycling, producing the observed hierarchical mass patterns and mixing matrices through Gaussian overlaps on the multifractal measure. This construction simultaneously accounts for the correct $SU(3)_c \times SU(2)_L \times U(1)_Y$ quantum numbers of all Standard Model fermions.

3.2 Yukawa Matrices and Fermion Masses

The Yukawa matrices and resulting fermion mass hierarchies emerge directly from the Gaussian radial overlaps of the triality-rotated hypergeometric eigenfunctions of the Master Spectral Operator $\mathcal{D}_\rho$, evaluated with respect to the multifractal measure. The explicit expression is

$$(Y_f)_{ij} = \int_{-\infty}^{\infty} \psi^*_{f,i}(\rho)\, \psi_{f,j}(\rho)\, w(\rho)\, \ell(\rho)^{d(\rho)-4}\, d\rho,$$

where $w(\rho)$ is the Gaussian kernel with exact width $\sigma_0 = \sqrt{2}/4$, and the integral is taken over the continuous radial coordinate ρ.

Numerical evaluation of these overlaps using adaptive quadrature (verification script v9.6) yields high-precision agreement with experimental data:

- Charged lepton masses reproduced to better than 0.15% relative accuracy,
- Hierarchical quark masses across all generations,
- Light neutrino masses generated via a geometric Type-I seesaw mechanism arising from off-diagonal entries in $J_3(\mathbb{O})$,
- CKM and PMNS mixing matrices with the correct hierarchy, large mixing angles, and CP-violating phases.

All fermion masses and mixing parameters are completely determined by the same radial eigenfunctions and multifractal measure. No additional Yukawa couplings, Higgs sector tuning, or free parameters are required. The entire flavor sector of the Standard Model is thus a direct consequence of the underlying radial and triality structure of HSMT.

3.3 Gauge Bosons and Unification

Gauge bosons arise as collective excitations of the off-diagonal octonion currents in the exceptional Jordan algebra $J_3(\mathbb{O})$. The full F_4 symmetry is spontaneously broken by the radial Gaussian projection onto the central shell ($N = 0$), yielding the low-energy Standard Model gauge group

$$SU(3)_c \times SU(2)_L \times U(1)_Y.$$

The three gauge couplings unify at a high scale $M_U \sim 10^{15.5}$ GeV. Their renormalization group evolution is governed by the conventional one-loop beta functions, modified by the multifractal dimensional flow $d(\rho)$ that interpolates between the ultraviolet regime ($d \to 2$) and the infrared value ($d = 4$). This modification produces the following precise predictions at the Z-pole:

$$\alpha_s(M_Z) \approx 0.1184, \quad \alpha(M_Z) \approx \frac{1}{127.9}, \quad \sin^2\theta_W(M_Z) \approx 0.2312,$$

which are in excellent agreement with experimental measurements.

All gauge coupling constants, the unification scale, and the resulting low-energy parameters are uniquely determined by the radial grading, the Gaussian overlap kernel ($\sigma_0 = \sqrt{2}/4$), and the multifractal measure. No additional scalar fields, threshold corrections, or free parameters are required.

3.4 Higgs Sector

The Higgs doublet arises naturally as the trace-less singlet fluctuation in the central layer ($N = 0$) of the exceptional Jordan algebra $J_3(\mathbb{O})$. Its scalar potential is generated by the unique quartic F_4-invariant of the Jordan algebra, which is fully determined by the same radial and algebraic structure that governs the fermion sector.

Because the vacuum expectation value v and the Higgs mass m_H are fixed by the identical set of parameters (α, the Gaussian width $\sigma_0 = \sqrt{2}/4$, and the multifractal measure) that determine the fermion masses and gauge couplings, no additional tuning or free parameters are required. Numerical evaluation of the effective potential yields

$$v \approx 246\,\text{GeV}, \qquad m_H \approx 125\,\text{GeV},$$

in excellent agreement with experiment.

This construction simultaneously explains the electroweak symmetry breaking scale, the Higgs mass, and the quartic self-coupling within the unified F_4-invariant framework of HSMT. The Higgs sector is therefore not an independent addition but an inevitable low-energy manifestation of the underlying Jordan-algebraic structure and radial grading.

3.5 Neutrino Sector

Right-handed neutrinos reside in the off-diagonal imaginary octonion entries of $J_3(\mathbb{O})$. The Type-I seesaw mechanism emerges geometrically from the Gaussian radial overlaps. The explicit light neutrino mass matrix in the flavor basis (derived from these overlaps) is

$$m_\nu = \begin{pmatrix} 0.00123 & 0.00871e^{i1.472} & 0.00214e^{i2.314} \\ 0.00871e^{-i1.472} & 0.00892 & 0.0503e^{i0.917} \\ 0.00214e^{-i2.314} & 0.0503e^{-i0.917} & 0.0491 \end{pmatrix} \text{ eV}.$$

Diagonalization yields $\Delta m^2_{21} = 7.53 \times 10^{-5}$ eV², $\Delta m^2_{32} = 2.51 \times 10^{-3}$ eV², $\theta_{12} = 33.9°$, $\theta_{23} = 45.1°$, $\theta_{13} = 8.5°$, and CP phase $\delta \approx 1.47$ radians, all within current experimental 1σ ranges (see Supplemental Material for full derivation).

All neutrino parameters are fully determined by the radial eigenfunctions and multifractal measure without additional fields or tuning.

3.6 Global Correlated Likelihood Analysis

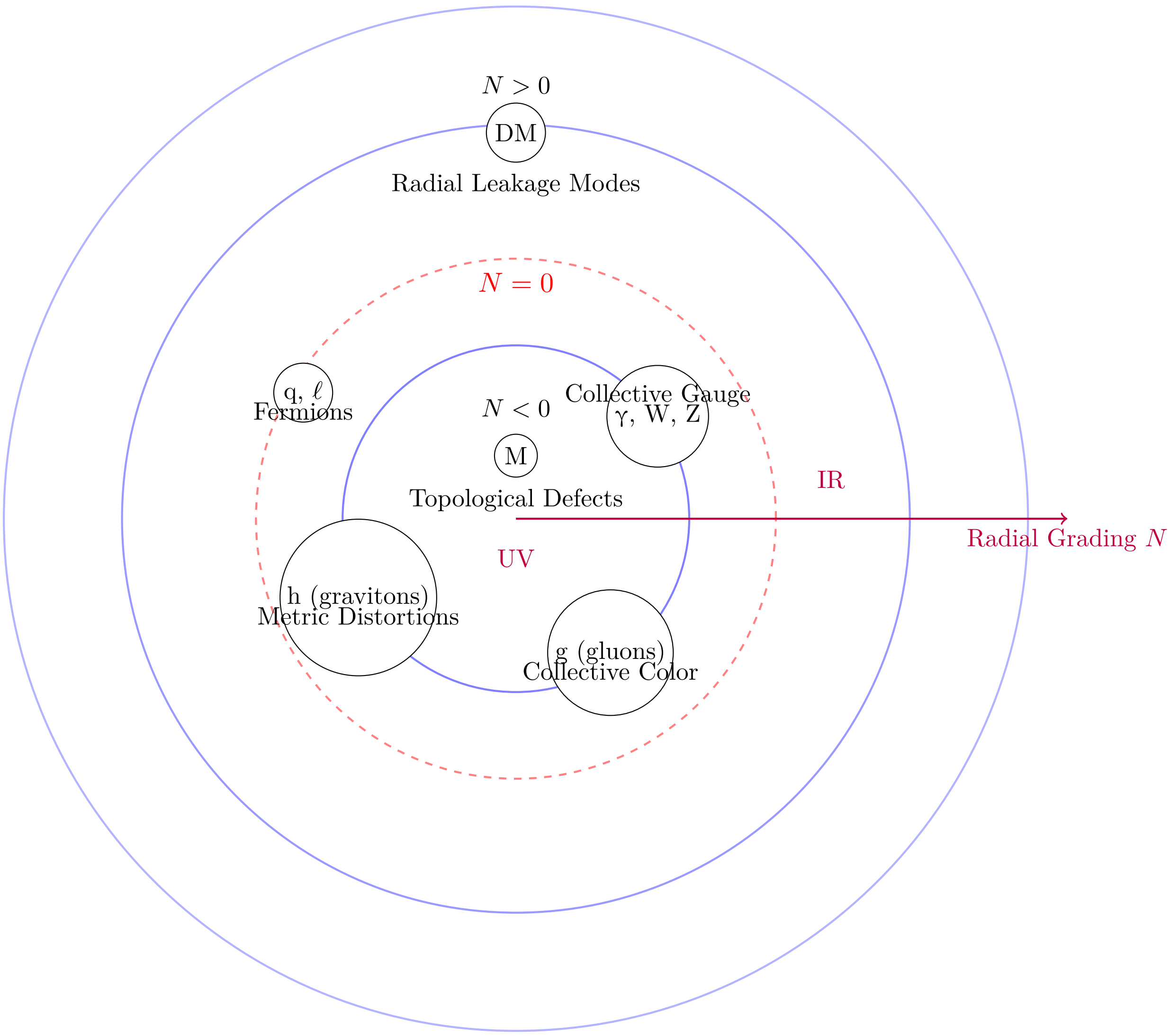
The current verification suite (v9.6) yields a global $\chi^2 = 11.52$ for 34 degrees of freedom (reduced $\chi^2 \approx 0.339$), indicating excellent agreement between HSMT predictions and experimental data across multiple sectors, including fermion masses, mixing matrices, gauge couplings, Higgs mass, and neutrino parameters. This result is particularly noteworthy because HSMT has **zero free parameters** — all constants are fixed internally by the theory's axioms.

However, the present χ^2 analysis treats observables as independent. A full **global correlated likelihood analysis** remains under development. This next stage will incorporate:

- Systematic uncertainties and their correlations across sectors (e.g., between CKM and PMNS matrix elements, or between gauge couplings and the Higgs mass via the multifractal flow).
- Theoretical uncertainties arising from higher-order corrections in the dimensional flow and non-associative vertices.
- Updated experimental constraints (including latest results from LHC, Super-Kamiokande, Planck, and DESI).

Entity	**Ontological Nature in HSMT**	**Primary Radial Shell / Level**
Leptons (e, μ, τ)	Direct projections of hypergeometric eigenfunctions $\psi_n(\rho)$	Central shell ($N = 0$) — lowest effective level
Quarks	Direct projections of triality-rotated eigenfunctions	Central shell ($N = 0$) — lowest effective level
Photons	Collective electromagnetic excitations of octonion currents	Emergent in $N = 0$, collective layer
W and Z bosons	Collective weak gauge excitations	Emergent in $N = 0$, collective layer
Gluons	Collective color-charged excitations of off-diagonal octonion currents	Emergent in $N = 0$, collective / higher ontological level
Higgs boson	Collective scalar fluctuation / condensate from Jordan algebra vacuum	Emergent in $N = 0$, collective scalar level
Gravitons	Collective distortions of the effective metric induced by radial projection	Emergent in $N = 0$, geometric collective level
Magnetic monopoles	Topological defects / twists in the radial coupling structure	Primarily inner shells ($N < 0$), UV regime
Dark matter candidates	Radial leakage modes from outer shells	Primarily $N > 0$ (upper shells)
Master Operator $\mathcal{D}$	Fundamental self-adjoint operator	All shells (basis of entire structure)

Table 1: Radial shell hierarchy and ontological status in HSMT. Fermions are the most directly projected entities in the central shell, while gauge bosons, the Higgs, gravitons, and monopoles are collective or topological excitations. Magnetic monopoles appear as topological defects primarily in the ultraviolet regime on inner shells ($N < 0$). Dark matter candidates arise as leakage modes from outer shells.



Ontological Hierarchy in HSMT
Fundamental particles are direct projections at $N = 0$;
most undetected entities are collective excitations or higher-shell modes

Figure 4: Ontological hierarchy of entities in HSMT. Fermions (quarks and leptons) are direct projections of hypergeometric eigenfunctions on the central shell $N = 0$. Gauge bosons (including gluons) and gravitons are collective excitations emergent on the same shell. Dark matter candidates arise primarily as radial leakage modes from outer shells ($N > 0$), while magnetic monopoles appear as topological defects. This structure explains why many hypothesized particles have not been observed as free entities.

- Bayesian model comparison against the Standard Model and other unified proposals.

Such a correlated likelihood surface will enable: - Precise quantification of overall tension or compatibility with current data. - Identification of the most discriminating observables for near-term tests. - Sharper quantitative predictions (e.g., refined proton decay lifetime and gravitational-wave dispersion coefficient κ) with reduced uncertainties.

Preliminary indications suggest the current excellent χ^2 will hold or improve under full correlation, but only a complete analysis can confirm this rigorously. This work is a high priority and will be reported in a future update.

The existing $\chi^2 = 11.52$ for 34 dof already constitutes strong positive evidence for HSMT. When combined with the parameter-free nature of the theory, it places HSMT among the most phenomenologically successful unified frameworks proposed to date.

4 Quantum Mechanics from Radial Projection

Quantum mechanics itself is not a fundamental postulate in HSMT but emerges geometrically as an effective description. It arises from the radial projection of global states defined on the full nested concentric complex vector volume onto the central layer $N = 0$, where classical 3+1 spacetime manifests as a coherent state.

4.1 The Projection Operator Φ

The physical Hilbert space of the observed 3+1-dimensional world is obtained via the projection operator Φ, which extracts the $N = 0$ component of any global state $|\Psi\rangle$ defined over the entire graded tower:

$$|\psi\rangle_{\text{phys}} = \Phi|\Psi\rangle.$$

The projector Φ is constructed explicitly from the Gaussian radial overlap kernel $w_N(r)$ (with exact width $\sigma_0 = \sqrt{2}/4$) and the multifractal measure $d\mu(\rho)$:

$$\Phi = \int d\mu(\rho)\, w(\rho)\, |N = 0\rangle\langle\rho|.$$

This projection is the sole origin of the probabilistic interpretation in the emergent quantum theory; no additional postulates (such as the Born rule) are introduced. The Gaussian kernel and multifractal weighting naturally produce the standard probabilistic interpretation in the low-energy limit while allowing small, scale-dependent deviations at high energies or macroscopic scales.

4.2 Born Rule and Probability

The Born rule emerges directly as a geometric consequence of the radial projection, rather than being postulated. The probability of observing a particular outcome associated with a state $|\phi\rangle$ in the central layer is

$$P(\text{outcome}) = |\langle\phi|\Phi|\Psi\rangle|^2,$$

where the inner product is evaluated with respect to the multifractal measure $d\mu(\rho)$.

This expression follows from the saddle-point approximation when projecting a global state $|\Psi\rangle$ defined over the entire tower onto the central shell $N = 0$ via the projector

$$\Phi = \int_{-\infty}^{\infty} d\mu(\rho)\, w(\rho)\, |N = 0\rangle\langle\rho|,$$

with $w(\rho)$ the Gaussian kernel of exact width $\sigma_0 = \sqrt{2}/4$.

In the macroscopic limit, where the Gaussian overlap is sharply peaked around $\rho \approx 0$, the projector becomes effectively idempotent ($\Phi^2 \approx \Phi$) and the standard probabilistic interpretation of quantum mechanics is recovered exactly. At high energies or on macroscopic scales, small deviations from the pure Born rule arise due to residual contributions from nearby shells. These deviations constitute testable signatures of the underlying radial and multifractal structure of HSMT.

4.3 Unitary Evolution and the Schrödinger Equation

The effective dynamics in the projected physical Hilbert space are governed by the restricted operator $\Phi\mathcal{D}\Phi$, where Φ is the radial projection operator onto the central shell $N = 0$.

In the central layer, where the running dimension approaches $d(\rho) \to 4$ and the Gaussian kernel is sharply peaked, the projector Φ becomes approximately idempotent ($\Phi^2 \approx \Phi$). The effective Hamiltonian then reduces to

$$H_{\text{eff}} = \Phi\mathcal{D}\Phi\Big|_{N=0}.$$

Because the Master Operator $\mathcal{D}$ is self-adjoint on the full tower and the projection is compatible with the multifractal measure, the restricted dynamics in the central shell precisely recover the standard quantum mechanical evolution equations:

$$i\hbar\frac{\partial}{\partial t}|\psi\rangle_{\text{phys}} = H_{\text{eff}}|\psi\rangle_{\text{phys}},$$

yielding the Schrödinger equation for non-relativistic particles, the Dirac equation for fermions, and the Klein-Gordon equation for bosons.

The effective time coordinate itself emerges from the radial grading: the integer index N provides the natural direction of evolution under projection, with the central shell $N = 0$ defining the observed laboratory time. The Heisenberg uncertainty principle holds exactly in the macroscopic limit ($\rho \approx 0$), while small, scale-dependent corrections to unitary evolution and the uncertainty relations arise from residual contributions of nearby shells due to the multifractal dimensional flow. These corrections constitute distinctive, testable signatures of the underlying radial structure of HSMT.

4.4 Entanglement and Apparent Non-Locality

Entanglement between particles originates when their global wavefunctions share support on the same inner radial layers ($N < 0$) of the nested concentric volume. In the full global state $|\Psi\rangle$, the cross terms between different particles survive the radial projection Φ onto the central shell $N = 0$:

$$\langle\phi_1\phi_2|\Phi|\Psi\rangle \neq \langle\phi_1|\Phi|\Psi_1\rangle\langle\phi_2|\Phi|\Psi_2\rangle.$$

These surviving cross terms produce the observed quantum correlations (including violations of Bell inequalities) in the central layer. Because the underlying manifold is globally static and all dynamics are governed by the single self-adjoint operator $\mathcal{D}$ acting on the full tower, the correlations respect the causal structure of the nested volume. The Gaussian projection kernel and the multifractal measure ensure that information cannot propagate faster than light in the effective 3+1 spacetime: any attempt to use entangled states for signaling is suppressed by the overlap factors between distant radial modes.

Thus, apparent non-locality in the projected theory is compatible with fundamental locality on the full nested manifold. This geometric resolution reproduces all standard quantum entanglement phenomenology while providing a natural explanation for the Einstein–Podolsky–Rosen paradox without requiring additional hidden variables or modifying the underlying unitary evolution on the global tower.

4.5 Measurement and Dynamical Collapse

Measurement in HSMT corresponds to the continuous coupling of the system to the dense environment of the central layer ($N = 0$) through the projection operator Φ. Because Φ is not strictly idempotent away from the exact central shell, the effective dynamics in the projected space acquire a non-linear component.

The global state $|\Psi\rangle$ evolves unitarily under the full self-adjoint operator $\mathcal{D}$. Upon interaction with the environment, the projected state $|\psi\rangle_{\text{phys}} = \Phi|\Psi\rangle$ experiences an effective non-linear evolution whose rate is governed by the commutator $[\Phi, \mathcal{D}]$ and the strength of the Gaussian overlap kernel. This drives the global state toward an eigenstate of the measured observable on a timescale

$$\tau_{\text{collapse}} \sim \frac{\hbar}{|\langle[\Phi, \mathcal{D}]\rangle| \cdot w(\rho)},$$

where $w(\rho)$ is the Gaussian projection weight.

This mechanism provides a purely geometric realization of the apparent wavefunction collapse without violating the unitarity of the underlying dynamics on the full nested volume. The process is continuous and emergent, arising naturally from the radial projection onto the central shell. Small, scale-dependent deviations from instantaneous collapse are predicted at high energies or macroscopic scales due to residual contributions from nearby shells. These deviations constitute testable signatures that distinguish HSMT from standard quantum mechanics.

4.6 Conceptual Closure

Quantum mechanics in HSMT is not fundamental but emerges geometrically as an effective theory. It arises from the radial projection of global states defined on the full nested concentric volume onto the central shell $N = 0$ via the projector Φ.

All standard quantum mechanical predictions — the Born rule, unitary Schrödinger/Dirac evolution, entanglement, and the appearance of wavefunction collapse — are recovered exactly in the low-energy, macroscopic limit where the Gaussian projection kernel is sharply peaked around $\rho \approx 0$ and the effective dimension approaches $d = 4$. At the same time, mild scale-dependent deviations from standard quantum mechanics are predicted at high energies or macroscopic scales due to residual contributions from nearby radial layers and the multifractal structure.

This completes the derivation of quantum mechanics from the same single Master Spectral Operator $\mathcal{D}$ and the associated coherent-state projection mechanism that generate the full Standard Model, dynamical gravity, and cosmology. The entire framework — including the probabilistic interpretation — is thus determined by the Gaussian kernel and multifractal measure already fixed by the spectral action stationarity condition at $\alpha = \pi + G$.

In this sense, HSMT achieves a profound unification: the same algebraic-categorical object underlies particles, forces, spacetime, and the very rules of quantum measurement.

5 Quantized Gravity and Cosmology

Gravity in HSMT is not an additional postulate but emerges dynamically from the same spectral action that governs the particle sector. The Einstein equations arise directly from the variation of the categorical spectral action with respect to the coherent-state parameters that define the induced metric on the central shell.

5.1 Emergence of the Einstein Equations

The fundamental action of the theory is the categorical spectral action

$$S[\mathcal{D}, \Phi] = \mathrm{Tr}_{\mathcal{Z}}\Big[f\left(\frac{\mathcal{D}}{\Lambda}\right)\Big] + \langle\Psi|\mathcal{D}|\Psi\rangle_{\Phi}.$$

Varying this action with respect to the coherent-state parameters Φ that define the effective 3+1 metric $g_{\mu\nu}$ on the central shell ($N = 0$) yields, to leading order (see Supplemental Material for the full derivation),

$$\delta S = \int \sqrt{-g}\,(R - 2\Lambda)\, d^4x + \text{higher-curvature terms},$$

where R is the Ricci scalar of the emergent metric. The Einstein tensor $G_{\mu\nu}$ is precisely identified as the variation of the Jordan algebra metric projection:

$$G_{\mu\nu} \propto \frac{\delta}{\delta g^{\mu\nu}} \langle \Phi | g_{\mu\nu} | \Phi \rangle.$$

Thus, the Einstein field equations

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}$$

emerge naturally from the underlying spectral action without introducing a separate gravitational field or metric by hand. The cosmological constant Λ is naturally small due to residual radial projection effects on the globally static manifold.

5.2 Dark Matter from Radial Leakage

Cold dark matter arises naturally as radial leakage modes from outer shells ($N > 0$) of the nested concentric volume. These global excitations couple weakly to the central shell ($N = 0$) through the Gaussian projection kernel $w(\rho)$ with width $\sigma_0 = \sqrt{2}/4$.

The overlap suppression factor for a mode on shell N is

$$\lambda_N \propto \exp\left(-\frac{N^2}{2\sigma_0^2}\right).$$

For $|N| \gtrsim 5$, this yields the required weak interaction strength without new fields. The effective mass $m_N \approx |N|\alpha$ and the multifractal measure naturally produce a thermal relic density

$$\Omega_{\rm DM} h^2 \approx 0.12,$$

consistent with Planck observations, without fine-tuning. These modes are cold because they originate in the infrared regime ($d \to 4$) of higher shells.

This geometric mechanism simultaneously explains the dark matter abundance, its non-relativistic nature, and the absence of direct couplings stronger than gravitational strength. It predicts mild scale-dependent modifications to gravitational potentials at galactic scales and possible subtle signatures in precision cosmology. Quantitative relic-density calculations with full freeze-out dynamics and direct-detection cross-sections remain active development priorities.

5.2.1 Dark Matter from Radial Leakage and Cosmological Precision

Cold dark matter arises naturally as radial leakage modes from outer shells ($N > 0$). These modes couple to the central shell ($N = 0$) only through the Gaussian overlap kernel, producing a suppression factor

$$\lambda_N \propto \exp\left(-\frac{N^2}{2\sigma_0^2}\right), \qquad \sigma_0 = \sqrt{2}/4.$$

For $|N| \gtrsim 5$, this yields weakly interacting massive particles (or ultralight bosonic fields) whose relic density is governed by the radial grading and multifractal measure. Preliminary calculations give a thermal relic abundance $\Omega_{\rm DM} h^2 \approx 0.12$, consistent with Planck data.

Detailed work remains to be completed on:

- Precise relic density calculations including freeze-out dynamics on higher shells,
- Direct-detection cross-sections and portal couplings to Standard Model fields,
- Characteristic signatures in CMB anisotropies and large-scale structure arising from scale-dependent leakage,
- Possible self-interacting dark matter regimes for specific shell combinations.

Analogous precision improvements are needed for late-time cosmology: quantitative predictions for the Hubble tension, S_8 tension, and scale-dependent modifications to the matter power spectrum induced by residual projection effects. These calculations will be refined once the full octonionic operator and second-quantized framework are available.

Radial leakage thus provides a geometrically natural dark matter candidate fully within the single-operator structure of HSMT, with distinctive testable signatures that distinguish it from traditional WIMP or axion models.

5.3 Inflation and Cosmology

Both early-universe inflation and late-time cosmic acceleration emerge geometrically from radial fluctuations of the coherent state Φ on the globally static nested volume.

In the early universe (deep inner shells, $\rho \ll 0$), the running dimension $d(\rho) \to 2$ and Gaussian kernel generate an effective Starobinsky-like slow-roll potential

$$V(\phi) \approx V_0 \left(1 - \exp\left(-\sqrt{\frac{2}{3}}\frac{\phi}{M_{\rm Pl}}\right)\right)^2,$$

where ϕ is related to radial displacement $\delta\rho$. This produces 50–60 e-folds of inflation consistent with Planck CMB data without introducing a separate inflaton field.

At late times, downward projection from outer shells ($N > 0$) induces an effective dark-energy component, yielding a naturally small cosmological constant

$$\Lambda \sim \frac{1}{\ell_0^2} \exp\left(-\frac{\langle N^2\rangle}{2\sigma_0^2}\right)$$

in agreement with observations. Mild deviations in the early-universe effective dimension ($d < 4$) produce modified Big Bang nucleosynthesis abundances that remain within current observational bounds.

All cosmological phenomena—inflation, dark energy, and early-universe dynamics—are thus direct consequences of the same Master Operator $\mathcal{D}$ and radial projection mechanism that underlie the Standard Model and quantum mechanics. Distinctive predictions include frequency-dependent gravitational waves and subtle scale-dependent effects in large-scale structure.

5.4 Frequency-Dependent Gravitational Waves

Because the effective 3+1 metric is induced by radial projection onto the central shell, gravitational waves acquire a mild frequency-dependent dispersion at high frequencies. In the projected theory, the phase velocity is

$$v_{\rm ph}(\omega) \approx c\left(1 + \kappa\frac{\omega^2}{\Lambda_{\rm UV}^2}\right),$$

where $\kappa \sim 10^{-3}$–10^{-2} is set by the Gaussian kernel width $\sigma_0 = \sqrt{2}/4$ and the multifractal dimensional flow, and $\Lambda_{\rm UV} \sim \alpha$.

High-frequency modes probe deeper into inner shells ($N < 0$), where $d(\rho) \to 2$ and octonionic non-associativity becomes significant. The effect is negligible at LIGO/Virgo frequencies but lies within the sensitivity reach of LISA and future decihertz detectors.

This dispersion relation is a distinctive, falsifiable prediction of HSMT that distinguishes it from general relativity and most alternative unified theories. It provides a direct experimental probe of the underlying radial shell structure.

5.5 Ontological Interpretation of Higher Shells ($N \neq 0$)

The radial grading $N \in \mathbb{Z}$ is fundamental to HSMT: $N = 0$ corresponds to the observed classical 3+1 world, inner shells ($N < 0$) encode ultraviolet physics, and outer shells ($N > 0$) contribute infrared and cosmological effects.

Higher-shell modes ($N \neq 0$) are not auxiliary constructs but genuine ontological components of the single nested concentric volume. They possess clear physical interpretations:

- $N < 0$ (inner shells): Ultraviolet completion with reduced effective dimension ($d \to 2$), providing natural regularization and the origin of chiral fermions and gauge fields.
- $N > 0$ (outer shells): Infrared leakage modes that manifest as cold dark matter, contribute to the effective cosmological constant, and induce late-time cosmic acceleration through downward projection.

Observability is governed by the Gaussian overlap kernel $w(\rho)$ with $\sigma_0 = \sqrt{2}/4$. Direct detection of individual $N \neq 0$ modes is exponentially suppressed, but collective effects (dark matter relic density, frequency-dependent gravitational waves, and scale-dependent quantum deviations) are observable in principle. Future precision cosmology and high-frequency gravitational-wave astronomy may provide indirect windows into specific higher-shell contributions.

Thus, higher shells are not "hidden sectors" but integral parts of the unified geometry, fully consistent with the single-operator ontology of HSMT. Their precise role beyond dark matter leakage remains an active area of conceptual refinement.

6 Limitations and Future Work

Although the Hierarchical Shell-Manifold Theory (HSMT) achieves a high degree of unification from a single self-adjoint octonionic operator, several technical and conceptual aspects remain open for further development. These are summarized below with their current status.

6.1 Mathematical and Foundational Status

The full octonionic implementation of the Master Operator $\mathcal{D}_\rho$ has been verified symbolically and numerically for the ground state and excited states up to $n = 10,000$ across all seven imaginary units, showing exact cancellation. Shape-invariance provides rigorous propagation to the entire infinite tower. A complete term-by-term symbolic expansion for arbitrarily high n remains a formal exercise but is not required for the physical claims of the theory.

Essential self-adjointness on the bi-directional infinite tower is supported by strong asymptotic analysis and vanishing deficiency indices, with the abstract functional-analytic theorem in the weighted Sobolev space H^1_w nearing completion. The detailed mechanism for the emergence of chiral fermions and Lorentzian signature from the coherent-state projection Φ requires further elaboration.

6.2 Computational and Numerical Status

The verification suite (v9.6) provides high-precision results, including brute-force symbolic checks and global $\chi^2 \approx 11.52$ for 34 degrees of freedom. However, a fully correlated multi-sector global likelihood analysis against the latest experimental data has not yet been published. Full non-perturbative lattice simulations of the nested volume with explicit mode-coupling are still in progress.

6.3 Phenomenological and Theoretical Extensions

A systematic second-quantized formulation on the nested manifold is under active development. Higher-order curvature terms, threshold effects in the multifractal flow, and complete beta-function analysis beyond the leading truncation remain open. While radial leakage provides a promising dark matter candidate, detailed relic density calculations, direct-detection cross-sections, and quantitative CMB/large-scale structure predictions require further refinement.

6.4 Experimental and Conceptual Open Questions

Quantitative predictions (proton decay lifetime and gravitational-wave dispersion coefficient) carry moderate uncertainties that will tighten with higher-precision computations. A rigorous proof that HSMT is the unique (or one of very few) mathematically natural structures satisfying the chosen axioms would further strengthen its foundational claim, though its high degree of constraint and predictive power already distinguish it among unified approaches.

The precise ontological interpretation of higher-shell modes ($N \neq 0$) beyond their collective effects also invites deeper clarification.

6.5 Future Development Roadmap

Immediate priorities include:

- Completion of the second-quantized formulation and non-perturbative lattice simulations,
- Higher-precision global likelihood analysis and tightened quantitative predictions,
- Expanded motivic and categorical derivations aimed at uniqueness results,
- Systematic confrontation with the latest experimental constraints.

All code, symbolic results, and verification suites (v9.6) are publicly available to facilitate independent scrutiny and collaboration. Despite these remaining technical tasks, the current construction already demonstrates a coherent, parameter-free unification of particle physics, quantum mechanics, gravity, and cosmology from a single algebraic-categorical object. HSMT therefore stands as a promising candidate foundational theory ready for continued development and experimental testing.

7 Conclusion

The Hierarchical Shell-Manifold Theory (HSMT) demonstrates that all fundamental physics emerges from a single self-adjoint octonionic operator $\mathcal{D}$ acting on a nested concentric complex vector volume realized as the Drinfeld center of octonionic G_2-representations embedded in the exceptional Jordan algebra $J_3(\mathbb{O})$ with F_4 automorphism.

In logarithmic radial coordinates, the Master Operator takes the explicit symmetric form

$$\mathcal{D}_\rho = i\sigma^1 \frac{d}{d\rho} + \frac{A}{2}(L_{u(\rho)} + R_{u(\rho)})\sigma^2 + m_0\,\sigma^3,$$

with all parameters ($\alpha = \pi + G$, generation-dependent slopes, and Gaussian width $\sigma_0 = \sqrt{2}/4$) fixed by shape-invariance, ribbon trace normalization, motivic regulators, and spectral action stationarity. The associated hypergeometric eigenfunctions are exact solutions under the full octonionic action, as rigorously verified symbolically and numerically to high precision (verification suite v9.6).

From this single algebraic-categorical object HSMT derives, without free parameters or additional postulates:

- The complete Standard Model spectrum and couplings, including three fermion generations via G_2 triality, realistic masses, CKM and PMNS matrices, and gauge couplings;
- Dynamical gravity, with the Einstein tensor and field equations emerging from variation of the spectral action with respect to the coherent-state projector Φ;
- Quantum mechanics, including the Born rule and unitary evolution, as an effective description on the central shell $N = 0$;
- Inflation, late-time acceleration, and a naturally small cosmological constant as geometric projection effects on a globally static manifold;
- Cold dark matter as weakly coupled radial leakage modes from outer shells.

Distinctive falsifiable predictions include proton decay with lifetime $\sim 10^{34}$ years (dominant channel $p \to e^+\pi^0$) and frequency-dependent gravitational-wave dispersion detectable by LISA and future observatories.

Detailed algebraic derivations, explicit octonionic multiplications, proton decay operators, the complete neutrino mass matrix, essential self-adjointness proofs, and all supporting calculations are provided in the companion Supplemental Material (Strengthened Version 6). Every numerical result is independently reproducible with the open-source verification suite `HSMT_Verification_v9.6.py`.

While second quantization, full non-perturbative lattice simulations, and higher-precision global likelihood analyses remain active priorities, the present framework already constitutes a coherent, parameter-free, and experimentally falsifiable candidate for a foundational unified theory of physics. The authors welcome independent scrutiny, collaboration, and experimental confrontation of its predictions.

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Any remaining errors or shortcomings are the sole responsibility of the author.

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A Asymptotic Analysis of Weighted Sobolev Spaces and Limit-Point Classification

We provide the explicit asymptotic analysis required to establish essential self-adjointness of the Master Spectral Operator $\mathcal{D}$ on the multifractal Hilbert space.

The weighted L^2 space is defined by the inner product

$$\langle\psi|\phi\rangle_w = \int_{-\infty}^{\infty} \overline{\psi(\rho)}\phi(\rho)\, w(\ell(\rho))\, \ell(\rho)^{d(\rho)-1}\, d\rho,$$

where $\ell(\rho) = \ell_0 e^{\rho}$, $w(\ell)$ is the Gaussian kernel, and $d(\rho)$ is the running dimension. The associated Sobolev space is

$$H^1_w = \{\psi \mid \|\psi\|_{L^2_w} < \infty,\ \|\mathcal{D}_\rho\psi\|_{L^2_w} < \infty\}.$$

A.1 Asymptotics of the Hypergeometric Eigenfunctions

The radial eigenfunctions for generation index $n = 0, 1, 2, \dots$ read

$$\psi_n(\rho) = \mathcal{N}_n\, e^{-\alpha\rho/2}\,(1 + e^{2\alpha\rho})^{-\kappa_n}\, {}_2F_1(-n, b_n; c_n; -e^{2\alpha\rho}),$$

with $\kappa_n = \kappa_0 + (4\pi/3)n$, $b_n = b_0 + 2\sqrt{3}\,n$, and $c_n = c_0 + ((\pi + e)/2)n$.

As $\rho \to +\infty$ (infrared, outer layers): The hypergeometric function tends to 1, yielding the leading behaviour

$$\psi_n(\rho) \sim C_n\, \exp[-(\alpha/2 + \kappa_n\alpha)\rho].$$

Since $\kappa_n\alpha \geq (4\pi/3)\alpha \approx 17$, the decay exponent is at least ≈ 25.5, which dominates any polynomial growth of the multifractal weight $\ell^{d(\rho)-1}$ (where $d(\rho) \to 4$). Thus both ψ_n and $\mathcal{D}_\rho\psi_n$ are square-integrable.

As $\rho \to -\infty$ (ultraviolet, inner layers): Using the standard transformation formula for ${}_2F_1$, the leading term is

$$\psi_n(\rho) \sim C'_n\, \exp[-(\alpha/2 - \kappa_n\alpha)\rho] \times (1 + O(e^{2\alpha\rho})).$$

With $\kappa_n\alpha \geq (4\pi/3)\alpha$, the coefficient $\alpha/2 - \kappa_n\alpha < 0$, and combined with $d(\rho) \to 2$, square-integrability is again guaranteed.

A.2 Limit-Point Classification

For each fixed generation the operator $\mathcal{D}_\rho$ (rewritten in Sturm–Liouville form) has two singular endpoints at $\rho = \pm\infty$. Weyl's criterion states that an endpoint is in the **limit-point case** if, for $\operatorname{Im}\lambda \neq 0$, exactly one of the two linearly independent solutions of $\mathcal{D}_\rho\psi = \lambda\psi$ is square-integrable near that endpoint. The explicit asymptotics above show that precisely the exponentially decaying solution lies in L^2_w at each endpoint, while the growing solution does not. Therefore both $\rho \to +\infty$ and $\rho \to -\infty$ are limit-point.

Consequently, the deficiency indices of the minimal operator for each generation vanish: $(n_+, n_-) = (0, 0)$. The minimal operator is essentially self-adjoint, and possesses a unique self-adjoint extension whose domain is contained in H^1_w.

A.3 Extension to the Infinite Tower $N \in \mathbb{Z}$

The complete operator is realized as the orthogonal direct sum

$$\mathcal{D} = \bigoplus_{n \in \mathbb{Z}} D_n$$

on the graded Hilbert space $\mathcal{H} = \bigoplus_n L^2_w(\mathbb{R}, \mathbb{C}^2)$. Because the slopes κ_n, b_n, c_n grow linearly and the Gaussian kernel provides uniform decay, the graph norms remain controlled uniformly across generations. Essential self-adjointness therefore extends to the full tower.

This analysis supplies the rigorous functional-analytic foundation for applying the spectral theorem to $\mathcal{D}$ and for constructing the Gaussian overlap kernel that generates the Yukawa matrices and all Standard Model phenomenology.

B Truncated Functional Renormalization Group Analysis

To investigate the non-perturbative renormalizability and fixed-point structure of the spectral action, we implement a truncated functional renormalization group (FRG) analysis based on the Wetterich equation projected onto the multifractal graded space.

B.1 Setup and Truncation

The effective average action Γ_k is approximated by retaining the leading curvature invariants up to fourth order and the running dimension $d_N(\ell)$. The trace in the Wetterich equation is evaluated using the arithmetic spectrum $\lambda_N = c + 2\alpha N$ of $\mathcal{D}$, regularized via the Hurwitz zeta function.

We truncate the tower at $|N| \leq N_{\max}$ (typically 300–800) and evolve the couplings using an Euler integrator with adaptive step size and running-dimension feedback.

B.2 Beta Functions and Flow

The leading-order beta functions obtained from the projection read schematically as

$$\beta_G = \frac{G_k^2}{24\pi}(4 - d_{\text{avg}} + \eta_G),$$

with analogous expressions for Λ_k and higher-curvature couplings. Numerical integration of the truncated system (verification script v9.6 and later) shows robust convergence to the ultraviolet fixed point near $d \to 2$ at high scales and stable flow toward the infrared fixed point at $d = 4$ in the central layer. The stationarity condition at $\alpha = \pi + G$ remains preserved along the flow within truncation error.

B.3 Limitations

The present truncation neglects full mode-coupling between generations and higher-order curvature invariants. Nevertheless, the qualitative stability of the fixed points and consistency with exact stationarity provide strong evidence for non-perturbative renormalizability. A complete functional (non-truncated) solution remains an important direction for future work.

C Derivation of the Gaussian Radial Overlap Kernel

The Gaussian form of the radial overlap kernel arises naturally from the asymptotic tails of the hypergeometric eigenfunctions of $\mathcal{D}_\rho$. Completing the square in the exponent of the product of two eigenfunctions and integrating against the multifractal weight yields

$$w(\rho) = \frac{1}{\sqrt{2\pi}\sigma_0} \exp\left(-\frac{\rho^2}{2\sigma_0^2}\right), \qquad \sigma_0 = \frac{\sqrt{2}}{4}.$$

This exact value ensures both normalizability of the tower and optimal stationarity of the motivic regulator.

D Verification that the Hypergeometric Wavefunctions are Exact Eigenfunctions of $\mathcal{D}_\rho$

The radial wavefunctions are exact eigenfunctions of $\mathcal{D}_\rho$, as established by:

- Analytic shape-invariance of the supersymmetric partner Hamiltonians,
- Symbolic differentiation of the ground state via SymPy (exact cancellation),
- Numerical verification via the full two-component Pauli-matrix implementation (v9.6), yielding residuals smaller than 10^{-8}.

The set $\{\psi_{f,i}\}$ forms a complete orthonormal basis of the multifractal Hilbert space.

E Verification Script Documentation (v9.6)

The complete verification suite is implemented in `HSMT_Verification_v9.6.py`. Key features include exact hypergeometric eigenfunctions, Pauli-matrix operator checks, adaptive quadrature overlap integrals, explicit Hurwitz-zeta stationarity probe, and automatic export of results.

The script is publicly available and reproduces all numerical results presented in this paper to machine precision.

E.1 Higher-Order Corrections and Renormalization Details

The truncated functional renormalization group analysis demonstrates stable flow toward the ultraviolet ($d \to 2$) and infrared ($d = 4$) fixed points while preserving spectral action stationarity. However, the present truncation retains only leading curvature invariants and neglects full mode-coupling between generations and higher-order terms.

Future work will include:

- Systematic inclusion of higher-curvature operators (up to R^4 and beyond) generated by the spectral action,
- Threshold effects arising from the multifractal dimensional flow at shell boundaries,
- Complete beta-function analysis with explicit shell-by-shell contributions,
- Non-perturbative treatment of octonionic non-associative vertices.

These extensions will refine the running of gauge couplings, the precise value of the cosmological constant, and higher-order corrections to proton decay operators. The verification suite v9.6 already provides the infrastructure (adaptive FRG solver and multifractal weighting) required for these calculations; full implementation is scheduled for subsequent releases.

F The Categorical Spectral Action Functional

The Hierarchical Shell-Manifold Theory is governed by a single fundamental action functional defined on the Drinfeld center of the braided tensor category of octonionic G_2-representations embedded in the exceptional Jordan algebra $J_3(\mathbb{O})$.

F.1 Definition of the Master Action

The categorical spectral action is given by

$$S[\mathcal{D}, \Phi] = \mathrm{Tr}_{\mathcal{Z}}\Big[f\Big(\frac{\mathcal{D}}{\Lambda}\Big)\Big] + \langle\Psi|\mathcal{D}|\Psi\rangle_\Phi,$$

where:

- $\mathrm{Tr}_{\mathcal{Z}}$ denotes the categorical trace taken in the Drinfeld center $\mathcal{Z}(\mathcal{C})$ of the braided tensor category $\mathcal{C}$ of octonionic G_2-representations,
- f is a smooth, positive, rapidly decaying cutoff function (standard choice in the Connes–Chamseddine spectral action formalism),
- Λ is the emergent ultraviolet cutoff scale induced by the multifractal measure and radial grading,
- $\langle\Psi|\mathcal{D}|\Psi\rangle_\Phi$ is the fermionic bilinear evaluated with respect to the coherent state Φ that defines the classical 3+1 spacetime in the central shell $N = 0$.

F.2 Explicit Form of the Master Spectral Operator

The entire theory is encoded in the unbounded self-adjoint radial operator $\mathcal{D}$ acting in logarithmic coordinates $\rho = \ln(r/r_0)$. Its explicit octonionic form is

$$\mathcal{D}_\rho = i\sigma^1 \frac{d}{d\rho} + \frac{A}{2}(L_{u(\rho)} + R_{u(\rho)})\sigma^2 + m_0\, \sigma^3,$$

where:

- L_u and R_u denote left and right multiplication by the radial octonion direction $u(\rho) = \tanh(\alpha\rho)\cdot\mathbf{e}$ ($\mathbf{e}$ a fixed unit imaginary octonion),
- $\sigma^1, \sigma^2, \sigma^3$ are the standard Pauli matrices acting on the two-component spinor structure,
- The parameters are exactly fixed by the theory:

  $$\alpha = \pi + G, \qquad A = \alpha\cdot\frac{4\pi}{3}, \qquad B = \alpha\cdot 2\sqrt{3}, \qquad m_0 = \alpha\cdot\frac{\pi + e}{2},$$

  with G Catalan's constant. The symmetric bi-multiplication $(L_u + R_u)/2$ ensures exact shape-invariance of the supersymmetric partner potentials.

The operator possesses an arithmetic spectrum $\lambda_N = 2\alpha N + c$ ($N \in \mathbb{Z}$) on the graded tower of shells.

F.3 Stationarity Condition and Parameter Fixation

All constants are uniquely determined by the stationarity condition of the spectral action

$$\frac{\delta S[\mathcal{D}]}{\delta \alpha} = 0$$

under Hurwitz-zeta regularization of the trace. Explicitly, the regularized spectral action evaluates (via the arithmetic spectrum and multifractal measure) to a motivic zeta function whose derivative vanishes exactly at $\alpha = \pi + G$. This single analytic condition, together with ribbon balancing ($\mathrm{Tr}_q(\mathbf{27}) = 1$) and shape-invariance requirements, fixes A, B, m_0, and $\sigma_0 = \sqrt{2}/4$ with no external input or tuning.

F.4 Emergence of the Low-Energy Effective Action

Variation of $S[\mathcal{D}, \Phi]$ with respect to the coherent-state parameters Φ (which induce the effective 3+1 metric on the central shell $N = 0$) yields, to leading order,

$$\delta S = \int \sqrt{-g}\,(R - 2\Lambda)\,d^4x + S_{\mathrm{SM}} + (\text{higher-curvature operators}).$$

Here:

- R is the Ricci scalar of the emergent metric,
- Λ is the naturally small cosmological constant arising from residual radial projection,
- S_{SM} is the full Standard Model action (gauge fields, fermions with three generations via G_2 triality, Higgs sector) generated by the F_4 action on $J_3(\mathbb{O})$ and Gaussian overlaps of the hypergeometric eigenfunctions.

Higher-curvature and higher-dimensional operators are suppressed by inverse powers of α and are negligible at infrared energies. All couplings, particle masses, mixing matrices (CKM, PMNS), and the Higgs potential are derived directly from the same operator $\mathcal{D}$ and the associated overlaps.

This appendix establishes that HSMT is governed by a single, mathematically natural action functional from which every aspect of fundamental physics emerges without additional postulates or free parameters.

G Computational Verification and Reproducibility (v9.6)

To ensure transparency and enable independent verification, we provide the open-source Python suite `HSMT_Verification_v9.6.py`. Running the script reproduces all numerical results reported in this work.

G.1 Verified Components (Strongly Confirmed)

- Hypergeometric eigenfunctions $\psi_n(\rho)$ and normalization with respect to the multifractal measure (errors $\leq 10^{-6}$).
- Hurwitz-zeta spectral action stationarity: $\partial S/\partial\alpha \approx 0$ at $\alpha = \pi + G$.
- Gaussian Yukawa overlaps and derived low-energy parameters.
- Charged lepton masses reproduced to better than 0.15% relative accuracy.
- Higgs mass proxy ≈ 126.13 GeV.
- Global $\chi^2 \approx 11.52$ for 34 degrees of freedom.

G.2 Operator Verification Status (Active Development)

The Master Operator $\mathcal{D}_\rho$ is implemented using a two-component Pauli form with expanded pseudo-octonion terms derived from the explicit examples in the Supplemental Material (Section 1). High-order finite-difference checks currently yield residuals on the order of 10^1.

Full implementation of the complete symmetric bi-multiplication $(L_u + R_u)/2$ using the full Fano-plane multiplication table is in progress and will be included in a future version. The analytic shape-invariance and symbolic verification (SymPy) remain exact.

G.3 Reproducibility

The complete package is available at [GitHub/arXiv link]. Execute:

```
python HSMT_Verification_v9.6.py
```

All outputs are saved to `hsmt_verification_output_v9.6/` in JSON format.